

* What is PL/SQL?

Ans- PL/SQL (Procedural Language / SQL) is Oracle's extension of SQL.

It combines SQL with programming constructs/features such as variables, loops, conditions, and exception handling, allowing complex business logic to be executed inside the database.

Ex:

SQL:-

Select * from Employee;

PL/SQL:-

Declare

v_name VARCHAR2(50);

BEGIN

SELECT name

INTO v_name

FROM employee

Where emp-id = 1;

DBMS_OUTPUT.PUT_LINE(v_name);

END;

* Why we use it?

Ans- SQL is used to work with data, while PL/SQL is used when we need programming logic such as IF-else statements, loops, variables and exception handling along with SQL ~~statements~~ statements.



* PL/SQL vs SQL :-

SQL	PL/SQL
①. Declarative language.	Procedural language.
②. No variables	Supports Variables.
③. No Loops	Supports Loops
④. No exception handling.	Supports exception handling.
⑤. Executes one statement at a time.	Executes a block of statements.
⑥. Used for ^{querying} querying data.	Used for business logic.

* Advantages of PL/SQL :-

- ①. Better Performance → A whole block ~~is~~ is sent to the database at once.
- ②. Error Handling
- ③. Supports Variables
- ④. Supports Loops
- ⑤. Code Reusability Using Procedures, Functions, Packages.



* Blocks of PL/SQL: - A PL/SQL program is written in blocks.

Parts of a PL/SQL Block: -

①. Declare Section: - Used to declare variables.

Ex: - DECLARE

V - salary NUMBER;

②. BEGIN section: - Contains executable statements.

Ex: - BEGIN

SELECT salary
INTO v - salary
from employee
where emp-id = 1;

END;

③. EXCEPTION section: - Handles errors.

Ex: Exception

when NO-DATA-FOUND THEN

DBMS-OUTPUT.PUT_LINE('no employee found');

* Types of PL/SQL Blocks: -

①. Anonymous Block: - No name.

②. Named Block: - Has a name.



* ~~Ques~~ Example of PL/SQL Block:-

DECLARE

v_name VARCHAR2(50);

BEGIN

SELECT name

INTO v_name

from employee

where emp-id = 1;

DBMS-OUTPUT.PUT_LINE(v_name);

EXCEPTION

WHEN NO-DATA-FOUND THEN

DBMS-OUTPUT.PUT_LINE('Employee not found');

END;

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